Journal of Sports Sciences

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/rjsp20

Force-velocity-power profiling of maximal effort sprinting, jumping and hip thrusting: Exploring the importance of force orientation specificity for assessing neuromuscular function

Nicklas Junge, Aske Lundsgaard, Mathias Foged Hansen, Pierre Samozino, Jean-Benoit Morin, Per Aagaard, Bret Contreras & Lars Nybo

To cite this article: Nicklas Junge, Aske Lundsgaard, Mathias Foged Hansen, Pierre Samozino, Jean-Benoit Morin, Per Aagaard, Bret Contreras & Lars Nybo (2021): Force-velocity-power profiling of maximal effort sprinting, jumping and hip thrusting: Exploring the importance of force orientation specificity for assessing neuromuscular function, Journal of Sports Sciences, DOI: 10.1080/02640414.2021.1920128

To link to this article: https://doi.org/10.1080/02640414.2021.1920128

Published online: 06 May 2021.

Submit your article to this journal

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=rjsp20

JOURNAL OF SPORTS SCIENCES
https://doi.org/10.1080/02640414.2021.1920128

Routledge
Taylor & Francis Group

SPORTS PERFORMANCE

Force-velocity-power profiling of maximal effort sprinting, jumping and hip thrusting: Exploring the importance of force orientation specificity for assessing neuromuscular function

Nicklas Junge[a,b], Aske Lundsgaard[b,c], Mathias Foged Hansen[a,c], Pierre Samozino[d], Jean-Benoit Morin[e,f], Per Aagaard[g], Bret Contreras[f] and Lars Nybo[a]

[a]Institute for Nutrition, Exercise and Sports, Department of Integrative Physiology, University of Copenhagen, Copenhagen, Denmark; [b]Stenhus Physical Test Center, Holbæk, Denmark; [c]Football Club, NorthZealand, Denmark; [d]Univ Savoie Mont Blanc, Inter-university Laboratory of Human Movement Biology, Chambéry, France; [e]Univ Lyon, UJM-Saint-Etienne, Inter-university Laboratory of Human Movement Biology, Saint-Etienne, France; [f]Sport Performance Research Institute, Auckland University of Technology, New Zealand; [g]Department of Sports Science and Clinical Biomechanics, University of Southern Denmark, Denmark

ABSTRACT

Comprehensive information regarding neuromuscular function, as assessed through force-velocity-power (FVP) profiling, is of importance for training optimization in athletes. However, neuromuscular function is highly task-specific, potentially governed by dissimilarity of the overall orientation of forceapplication. The hip thrust (HT) exercise is thought to be of relevance for sprinting considering its antero-posterior force orientation and considerable hip-extensor recruitment, however, the association between their respective FVP profiles remains unexplored. Therefore, to address the concept of force orientation specificity within FVP profiling, the maximal theoretical neuromuscular capabilities of 41 professional male footballers (22.1 ± 4.1 years, 181.8 ± 6.4 cm, 76.4 ± 5.5 kg) were assessed during sprint acceleration, squat jumping (SJ) and the HT exercise. No significant associations were observed for maximal theoretical force or velocity between the three FVP profiling modalities, however, maximal theoretical power (Pmax) was correlated between sprinting and SJ ($r = 0.73$, $P < 0.001$) and HT and SJ ($r = 0.44$, $P = 0.01$), but not between sprinting and HT ($r = 0.18$, $P = 0.36$). In conclusion, although Pmax may be considered a somewhat universal lower-extremity capability, neuromuscular function is associated with substantial task-specificity not solely governed by the overall direction of force orientation.

ARTICLE HISTORY
Accepted 16 April 2021

KEYWORDS
Neuromuscular; force-vector; hip thrust; sprint; squat jump

INTRODUCTION

Maximal intensity brief actions such as sprint accelerations, changes of direction and vertical jumps often immediately precede goals and scoring opportunities in elite level football (Faude et al., 2012), highlighting the importance of ballistic proficiencies within the players(Cometti et al., 2001). The performance level in such actions is predominantly governed by the maximal velocity-specific force production capabilities of the neuromuscular system (Marcote-Pequeno et al., 2019; Morin et al., 2019; Samozino et al., 2014), characterized by maximal theoretical force- (F0), velocity- (V0) and power (Pmax), which are confined within the individual boundaries of the linear force-velocity (F-v) and parabolic power–velocity relationships, respectively. Although traditional performance measures such as sprint times and jump heights are indeed of importance, especially for monitoring longitudinal adaptations, they are of limited value towards individualized training optimization, as information regarding the underlying neuromuscular capabilities and the balance or imbalance herein remain unknown. To solve this problem, the concept of force-velocity-power (FVP) profiling has received increasing scientific attention due to its provision of comprehensive information regarding the neuromuscular capabilities underpinning ballistic or explosive-type movement performances.

Although FVP profiling has been extended to numerous movement patterns/tasks(Alonso-Aubin et al., 2019; Garcia-Ramos et al., 2016; Padulo et al., 2017; Rahmani et al., 2018; Sreckovic et al., 2015), maximal vertical jumping (Samozino et al., 2008) and horizontal sprint accelerations (Samozino et al., 2016) appear to be the predominant modes of application. Previous studies have reported a marked distinctiveness of the neuromuscular capabilities in the horizontal and vertical planes(Escobar-Alvarez et al., 2020; Jimenez-Reyes et al., 2018; Marcote-Pequeno et al., 2019), thus creating the need to include both profiling modalities in a comprehensive multi-planar neuromuscular "profile". Although both tasks (vertical jumping, horizontal sprint acceleration) are essentially closed kinetic chain triple-joint extension movements, a likely fundamental reason for this apparent distinctiveness is the dissimilarity of the overall movement- and thus external force orientation in reference to the longitudinal axis of the body between the two tasks that results in different coordinatory demands, joint-angle ROMs, muscle length changes and muscle recruitment patterns.

CONTACT Nicklas Junge nicklas.junge@nexs.ku.dk Dronning Dagmars Vej 78, 3650 Olstykke, Denmark.

© 2021 Informa UK Limited, trading as Taylor & Francis Group

The hip thrust (HT) exercise has received increasing attention due to its high level of hip-extensor activation and sagittally directed force orientation, both of which are theoretically well-suited for improving the primary mechanical determinant of sprint performance, i.e. the horizontal ground reaction force (Morin et al., 2011; Rabita et al., 2015). This notion is supported by the effectiveness of this exercise compared to axially loaded exercises for augmenting sprinting ability(Abade et al., 2019; Contreras et al., 2017), although not a universal finding(Gonzalez-Garcia et al., 2019; Jarvis et al., 2019). Cross-sectionally, moderate to nearly perfect relationships between HT peak force (Williams et al., 2021) and mean power(Loturco et al., 2018) and sprinting ability have been reported; however, to our best knowledge, no previous studies have utilized the FVP profiling concept to assess HT-specific neuromuscular function while concurrently evaluating its relationship to sprint ability or sprint-specific neuromuscular function. Thus, it is currently unknown to which extent the F-v relationships of maximal effort sprinting and hip thrusting are inter-related. Importantly, knowledge about this possible relationship has potential implications for the design of athlete testing and training practices, respectively.

Therefore, the purpose of the present study was to examine the relationships between the maximal theoretical neuromuscular capabilities of vertical squat jumping, horizontal sprinting and the HT exercise, respectively, to gain insight into the assessment of neuromuscular function through FVP profiling. Based on the literature, it was hypothesized that absent or only weak relationships would be observed between jumping (squat jump, SJ) and sprinting F0, V0 and Pmax, whereas moderate- to high-magnitude associations would be observed between HT and sprinting FVP parameters, especially for V0, considering the strong influence of hip-extensor activation on late-phase sprint acceleration capacity.

MATERIALS AND METHODS

Experimental overview

The present study consisted of three experimental sessions, comprising FVP profiling for the horizontal sprint acceleration, HT exercise and vertical squat jumping, respectively. Sessions were separated by at least 48 hours and conducted within a one-month period during the late preseason and start of the competitive season for both teams (August–September 2020). A thorough warm-up was performed involving both general (running drills for sprint acceleration, cycling for HT and SJ) and specific elements, in which the respective motor tasks were performed with progressively increasing effort.

Participants

Using convenience sampling, 41 professional male footballers (22.1 ± 4.1 years, 181.8 ± 6.4 cm, 76.4 ± 5.5 kg, 3 goal keepers, 13 defenders, 12 mid-fielders and 13 attackers) from two teams competing in the premier Danish division (Danish "Super League") were recruited for the present study. Due to various circumstances (player transfers, minor injuries, etc.), not all participants completed the full test-battery, resulting n = 29, 26 and 32 completing both the sprint/HT, sprint/SJ and HT/SJ, respectively. All participants were used to maximal effort sprinting and resistance training; the latter including both the HT and SJ exercises utilizing heavy (80% 1-RM) external loads. Thus, no specific familiarization was conducted aside from the warm-up procedures performed immediately prior to testing. Participants habitually underwent 5–6 football practice sessions and competed in 1–2 tournament matches per week. At the time of testing all participants were free of injuries, not taking dietary supplements with ergogenic properties and were instructed to refrain from performing intense physical activity on the day preceding testing and to abstain from consumption of alcohol and caffeine for 12 hours prior to commencement of the experimental procedures. Thorough information regarding potential risks and discomforts associated with the experimental procedures were given to the participants before they signed the consent form. The study was approved by the local ethical committee and conformed to the Declaration of Helsinki.

Sprint acceleration FVP profiling

Sprint acceleration FVP profiling was conducted according to the macroscopic inverse dynamics approach of Samozino et al.(Samozino et al., 2016). Three maximal effort 30-m sprint accelerations were conducted from a standing start interspersed by 5 min of recovery, with split times recorded by single-beam photocell timing gates (Witty Sprint System, Microgate, Italy) positioned alongside the track at 2.5, 5, 10, 15, 20, 25 and 30 m at a height corresponding roughly to the participants' centre of mass (1 m). Mean split times of the two fastest attempts were calculated and utilized for the computations as previously recommended(Haugen et al., 2020). A modified pair of timing gates placed at ground level on either side of the back foot of theparticipants allowed for initiation of timing as soon as movement occurred during the initial step-cycle. Using a spreadsheet originally developed by Morin and Samozino (2017) and subsequently modified by Stenroth et al. 2020) (https://www.researchgate.net/publication/342233454_Spreadsheet_for_sprint_acceleration_force-velocity-power_profiling_with_optimization_to_correct_start_time), the time-position data were fitted to a mono-exponential time-velocity function via a least squares methodology through optimization of three variables using the Excel solver function; 1) Vmax, 2) acceleration time constant and 3) time-shift, the latter variable previously utilized by (Morin et al., 2019) and subsequently validated by (Stenroth et al., 2020). Step-averaged ground reaction forces and its horizontal (F_H) and vertical (F_V) components were computed as a function of running velocity based on instantaneous acceleration, body mass and aerodynamic drag, the latter being estimated from frontal surface area

of the individual participant, ambient conditions and instantaneous running velocity. Sprint-specific maximal theoretical neuromuscular capabilities were derived through extrapolation of the linear F_H-velocity function to its x and y intercepts, F0 being the extrapolated force at null velocity (normalized to body mass), V0 the extrapolated velocity at null force, SFv the regression slope and Pmax = F0×V0/4 Jaric (2015). Sprint testing was conducted on an outdoor (17.0 ± 2.8°C, 1018 ± 6 hPa) artificial grass field with subjects wearing football cleats. For a comprehensive description of the methodology and associated computations, the reader is referred to the original work of Samozino et al.(Samozino et al., 2016).

Hip thrust FVP profiling

The maximal neuromuscular capabilities of the HT movement were quantified through maximal effort HT attempts performed against external loads of +40, +60, +80, +100 and +120 kg, applied with a free Olympic barbell equipped with a thick (d = 10 cm) padding to minimize participant discomfort. The HT kinematics were in accordance with the recommendations of Contreras et al.(Contreras et al., 2011, 2015), i.e. with the upper back resting on the bench, feet positioned shoulder width apart and hip- and knee-joint angles of ~ 90 deg, the movement was executed predominantly through extension of the hips, causing the pelvis and barbell to ascend vertically. Participants were instructed to perform the lift with maximal intentional velocity throughout the full range of motion, thus ballistically projecting the barbell off their hips if permitted by the load. A bench specifically designed for this exercise (LMX1057 Crossmax Hip Thrust Bench, height of 40 cm) positioned within a power rack was utilized, with the barbell resting on mechanical stops at a height of 30 cm. Subjects were instructed to reduce system compliance and impact forces as much as possible by pushing their hips into the barbell padding prior to the lift. Three attempts at each load were conducted interspersed by a brief ~ 10–20 s rest and 3 min of recovery separating each loading sequence. To quantify the duration of the concentric lifting phase and subsequently compute mean concentric velocity, high-frequency (240 Hz) video analysis was performed with MyLift v.10.0.1(Balsalobre-Fernandez et al., 2017) by manually selecting the initial and final frames of the lift, corresponding to the first instant of vertical displacement of the barbell and attainment of a parallel body position, respectively. The latter was cross-referenced to an a priori obtained still image, using coloured background markings to facilitate identification of the terminal position as objectively as possible. Both videos and still images were captured using a tripod-mounted iPhone XR positioned in parallel to the longitudinal axis of the barbell at a distance of 1.5 m. To quantify and account for the total mass of the moving system (Rambaud et al., 2008), which consists of both the barbell and a fraction of body mass, subjects were weighed on a force plate (Pasport force platform, 1 KHz, PASCO, Denmark) positioned beneath the gluteal region while resting in the middle of the range of motion. Total system mass used in the following computations thus accounts for both external (barbell + weight plates) and internal (body) load.

Using the attempt with the highest mean concentric velocity of each loading condition, the following computations were performed using fundamental laws of dynamics:

$$a = \frac{2d}{t^2} \tag{1}$$

$$F = m * (a + g) \tag{2}$$

Where t is the duration of the concentric phase, d is the vertical displacement of the barbell, a is mean concentric acceleration, m is the total system mass and g is the gravitational acceleration (9.81 m/s^2).

Individual F-v relationships were constructed through application of least squares linear regression to the computed force and velocity data from the five loading conditions. Extrapolation of the linear regression functions to the force (normalized to body mass) and velocity axes yielded F0 (N/kg) and V0 (m/s), respectively, with SFv (N/kg/m/s) defined as the regression slope and Pmax = F0×V0/4 Jaric (2015).

Squat jump FVP profiling

Maximal theoretical neuromuscular capabilities for squat jumping were quantified according to Samozino et al. (Samozino et al., 2008). Height of push-off (H_{PO}) was a priori determined as the difference between lower-extremity length and the greater trochanter to ground distance in the lowermost SJ position. Hereafter, maximal effort SJs were performed unloaded and against external loads of +30 and +60 kg, three attempts being performed against each loading condition, with a recovery period of 3 min interspersing each jump. Loading was applied via a plate loaded Olympic barbell in the externally loaded jumps, whereas an 0.4 kg wooden bar was utilized for the unloaded jumps to facilitate kinematical consistency throughout. Subjects performed a controlled descent to an individually predetermined position of ~90 deg of knee flexion, maintained this position for 2 s and subsequently executed a maximal effort concentric jump devoid of countermovement.

For the highest jump of each loading condition, mean concentric force and velocity were computed through an inverse dynamics approach based on total system mass, H_{PO} and vertical jump height, the latter estimated from aerial time using a validated optical system (OptoJump, Microgate, Italy)(Glatthorn et al., 2011). FVP parameters were subsequently derived through extrapolation of the linear F-v function (computed from the three loading conditions) in a manner consistent with the HT. Detailed description of the methodology and associated computations are reported in Samozino et al.(Samozino et al., 2008).

Statistical analyses

The relationships between the same FVP profile parameters, e.g., F0 vs. F0, V0 vs. V0 etc., of the sprint acceleration, HT and SJ were quantified through Pearson's product moment correlation obtained via simple linear regression. Correlation coefficients (r) of <0.2, 0.2–0.5, 0.5–0.8 and >0.8 were defined as correlations of negligible, low, moderate and high strength, respectively. P-values <0.05 were considered statistically significant. Values are reported as group mean ± standard deviation. All statistical analyses were performed using GraphPad Prism 8.3.0.

RESULTS

Force and velocity data of both the HT (r^2 = 0.94 ± 0.08) and SJ (r^2 = 0.97 ± 0.04) were fitted by least squares linear regression while the sprint acceleration time-velocity data were fitted by a mono-exponential function (r^2 = >0.99 ± <0.1). Mean F0 was 7.6 ± 0.9, 42.9 ± 11.1 and 37.8 ± 4.8 N/kg for the sprint, HT and SJ, respectively, with accompanying V0 values of 9.82 ± 0.48, 2.83 ± 1.05 and 3.28 ± 0.60 m/s, resulting in a Pmax of 18.6 ± 2.2, 29.4 ± 8.2 and 30.6 ± 4.7 W/kg. A representative sprint acceleration time-velocity function and derived F-v relationship are illustrated in Figure 1, while representative SJ and HT F-v relationships are illustrated in Figure 2. Prior to time-shifting (+0.32 ± 0.09 s), 30-m sprint time was 4.08 ± 0.13 s with unloaded SJ height being 40.4 ± 4.7 cm. No significant correlations were observed for F0, V0 and SFv between the three FVP profiling modalities (r = −0.19–0.20, P ≥ 0.27). In contrast, positive correlations were observed for Pmax between the sprint and SJ (r = 0.73, P < 0.001) and between the SJ and HT (r = 0.44, P = 0.01), while absent between sprint and HT (r = 0.18, P = 0.36). All correlation coefficients and associated 95% CIs and P-values between the three profiling modalities are reported in Table 1, Table 2, Table 3. Relationships for Pmax calculated between sprint and SJ and sprint and HT are illustrated in Figures 3 and Figure 4, respectively.

DISCUSSION

Main findings

The main observation in the present study was the absence of relationships between force production capacities during the HT exercise and sprint acceleration, whatever the velocity, which was unexpected given the high hip-extensor dependency and large magnitude of sagittally directed force production involved in both motor tasks. Sprint-specific neuromuscular function thus appears as a complex multifactorial phenomenon not solely governed by "isolated" hip-extensor capabilities. An ancillary but noteworthy finding was the moderate relationship observed for Pmax between the SJ and sprint acceleration, however, without accompanying associations between the underlying mechanical capabilities of F0 and V0. Thus, the present data highlight, in trained individuals, the importance of movement/task specificity for the assessment of neuromuscular function with both force orientation specificity and dynamic correspondence being of likely importance.

Inter-relationships between the SJ, HT and sprint

In the present study a moderate association was observed between Pmax of sprinting and vertical squat jumping, however, no accompanying relationships were noted between its underlying mechanical determinants, i.e. F0 and V0. The present findings corroborate, at least in part, previous reports by Marcote Pequeño et al.(Marcote-Pequeno et al., 2019) and Jimenez-Reyes et al.(Jimenez-Reyes et al., 2018), who reported absent to poor correlations between F0 and V0 of sprinting and jumping in elite footballers, however, with conflicting results regarding Pmax, which was associated in the former study (r = 0.75) but not the latter (r = 0.15). The apparent FVP profile distinctiveness of these two motor tasks as observed in the present study is likely attributable in part to the elite level of the subjects, as Jimenez-Reyes et al.(Jimenez-Reyes et al., 2018) reported a progressive associative deterioration between horizontally- and vertically oriented neuromuscular capabilities as a function of subject level of practice. The modest correspondence between these factors as observed in the present study and also previously reported by others(Jimenez-Reyes et al., 2018; Marcote-Pequeno et al., 2019), is logical considering the discrepant biomechanics of the two tasks (sprint vs. SJ). Although both are heavily dependent upon lower-extremity triple extension segmental dynamics, the discrepant force

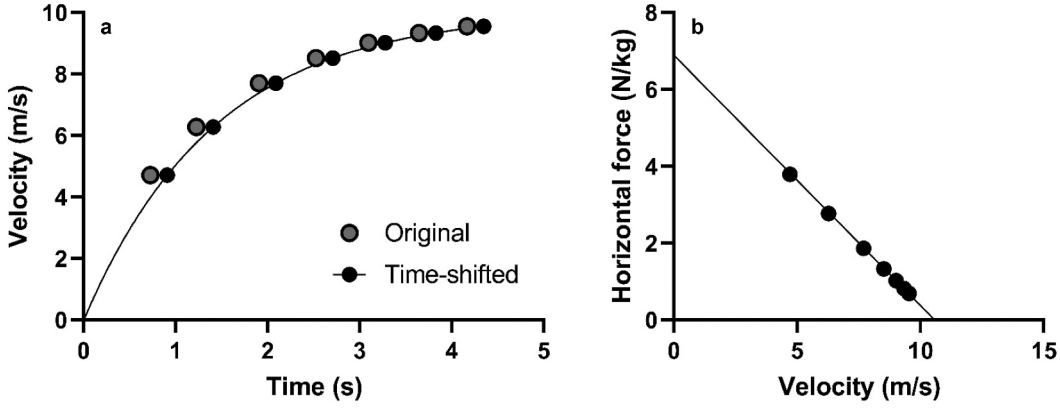

Figure 1. Panel A: exponential time-velocity function of a representative subject with a time-shift of 0.183 s, an F0 of 6.89N/kg, V0 of 10.59 m/s, Pmax of 18.24 W/kg and acceleration time constant of 1.44 s. The grey circles illustrate the original time-velocity data points derived from simple split times whereas the black circles and associated exponential function are derivatives of the time-shift methodology(Stenroth et al., 2020). Panel B: body mass-normalized horizontal force production as a function of sprinting velocity computed from the time-velocity function according to the methodology of Samozino et al.(Samozino et al., 2016), with the black circles representative of the same time points as in panel A.

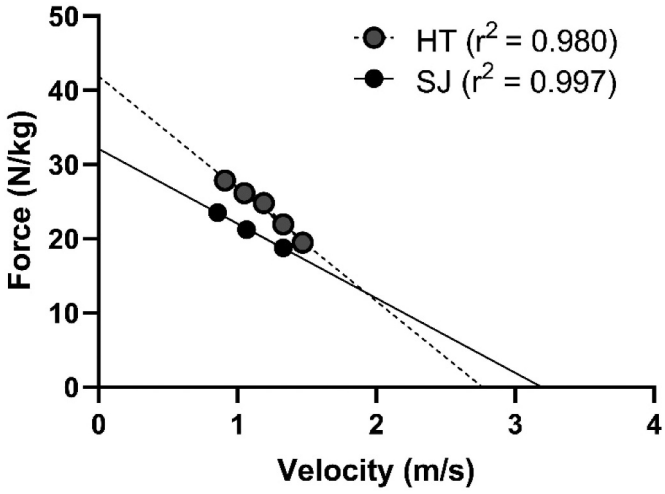

Figure 2. Illustration of the force–velocity relationships of the hip thrust (HT [grey circles and dashed line]) and squat jump (SJ [black circles and solid line]) of a representative subject. The force-velocity data points of the SJ are computed according to the inverse dynamics approach of Samozino et al.(Samozino et al., 2008) whereas the HT data points are products of double mathematical integration using high-frequency video analysis(Balsalobre-Fernandez et al., 2017; 2018).

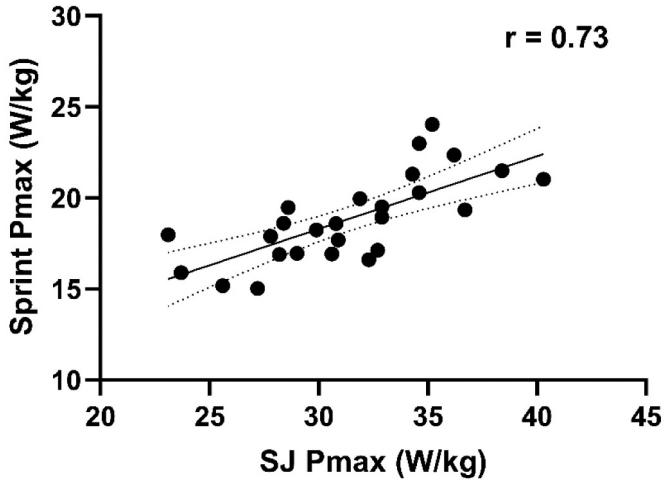

Figure 3. Illustration of the moderate relationship (P < 0.001) and associated 95% confidence intervals (0.47: 0.87) between the maximal theoretical power (Pmax) of the squat jump (SJ) and sprint acceleration (sprint), both in watts per kilo of body mass (W/kg).

Table 1. Pearson's correlation coefficients (r-value), 95% confidence intervals (95% CI) and P-values for the relationships between the force-velocity-power profiling parameters of the sprint acceleration (sprint) and squat jump (SJ).

		F0	V0	Pmax	SFv
Sprint/SJ	**r-value**	0.17	−0.07	**0.73**	−0.17
(n = 26)	**95% CI**	(−0.24: 0.52)	(−0.45: 0.32)	(0.47: 0.87)	(−0.53: 0.23)
	P-value	0.42	0.72	< 0.001	0.40

Table 2. Pearson's correlation coefficients (r-value), 95% confidence intervals (95% CI) and P-values for the relationships between the force-velocity-power profiling parameters of the sprint acceleration (sprint) and hip thrust (HT).

		F0	V0	Pmax	SFv
Sprint/HT	**r-value**	0.01	−0.15	0.18	−0.19
(n = 29)	**95% CI**	(−0.36: 0.37)	(−0.49: 0.23)	(−0.20: 0.51)	(−0.52: 0.19)
	P-value	0.98	0.43	0.36	0.31

Table 3. Pearson's correlation coefficients (r-value), 95% confidence intervals (95% CI) and P-values for the relationships between the force-velocity-power profiling parameters of the squat jump (SJ) and hip thrust (HT).

		F0	V0	Pmax	SFv
SJ/HT	**r-value**	0.20	0.02	**0.44**	−0.02
(n = 32)	**95% CI**	(−0.16: 0.51)	(−0.33: 0.37)	(0.11: 0.68)	(−0.37: 0.33)
	P-value	0.27	0.92	0.01	0.91

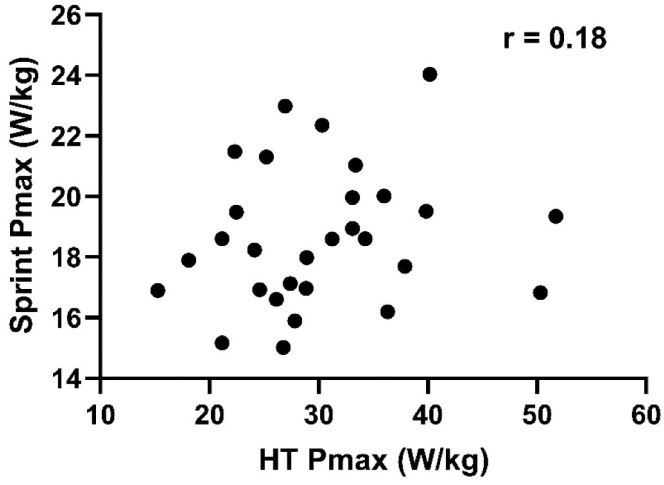

Figure 4. Illustration of the non-significant (P = 0.36) relationship between the maximal theoretical power (Pmax) of the hip thrust (HT) and sprint acceleration (sprint), both in watts per kilo of body mass (W/kg).

orientations are likely to require quite dissimilar muscle recruitment patterns, where sprinting is a hip-extension dominant task while knee extension capabilities are of relatively greater importance for the vertically oriented ground reaction forces in maximal vertical jumping(Morin et al., 2015; Wiemann & Tidow, 1995). However, although concerned with overall perpendicularly oriented forces, the inclined posture adopted during early acceleration allows for a component of the "local" force, i.e. in reference to the overall body orientation, to be transmitted horizontally onto the global supporting ground, which could explain the pattern of quadriceps activation as a function of sprinting velocity, i.e. high initial recruitment during early acceleration and subsequent attenuation at higher velocities (Girard et al., 2020; Morin et al., 2015). The kinematic dependence upon horizontal velocity could underlie the greater, although still insignificant, correlation between F0 than V0 as observed in the present study and by Jimenez-Reyes et al. (Jimenez-Reyes et al., 2018), but not by Marcote Pequeño el al.(Marcote-Pequeno et al., 2019), and could also be a contributory factor towards the relatively robust correlation of Pmax, as this mechanical parameter is realized during early acceleration due to the exponentiality of the time-velocity function and thus rapid attainment of the optimal velocity.

In light of previous reports regarding the distinctiveness of vertical and horizontal FVP profiles(Jimenez-Reyes et al., 2018; Marcote-Pequeno et al., 2019) and the apparent connection between HT performance and sprint acceleration capacity (Loturco et al., 2018; Williams et al., 2021), the present study sought to elucidate whether this connection extends towards similar neuromuscular capabilities, thus additionally exploring

the importance of force orientation specificity for assessment of neuromuscular function. In contrast to Loturco et al.(Loturco et al., 2018), who reported strong-to-excellent relationships between HT power output and sprint velocities over distances ranging from 10 to 150 m and Williams et al.(Williams et al., 2021), who observed a moderate relationship between peak HT force and peak sprint velocity, no such relationships could be observed in the present study. Several methodological discrepancies between these studies could explain the contrasting observations. In the Loturco study(Loturco et al., 2018), HT ability was quantified as the maximal mean propulsive power observed during an incremental loading test and the peak HT force of Williams et al. (Williams et al., 2021) was obtained during 3RM loading, both load-specific measures not directly representative of the underlying neuromuscular function. However, it should be noted that various post-hoc analyses of the present data in terms of load-specific HT kinetics and simple sprint split times or maximal running velocity (data not reported) did not reveal any statistically- or practically meaningful relationships, suggesting it to be an unlikely source of dissimilarity. Methodological differences pertaining to the quantification of HT kinetics, high-frequency video analysis vs. linear position transducer (Loturco et al., 2018) vs. force platform (Williams et al., 2021), and range of motion considered, full range of motion vs. considering only the propulsive phase (Loturco et al., 2018) vs. a joint-angle specific measure (Williams et al., 2021), could potentially underlie our contrasting findings. That being said, the most likely methodological source of disparity between the studies is the heterogeneity and practice level of the participant population, as Loturco et al. (Loturco et al., 2018) encompassed both male and female sprinters and jumpers and Williams et al. (Williams et al., 2021) included various male team sport players, whereas only male professional footballers were included in the present study. This notion is supported by the higher coefficients of variation (CV) in Loturco et al.(Loturco et al., 2018) for both HT power output (27.7 vs. 11.3–16.3%) and sprint ability (5.3–8.9 vs. 3.3–4.9%), with CVs not extractable from Williams et al. (Williams et al., 2021), and it could thus be that the kinematic dissimilarities between the two tasks are pronounced enough to govern an apparent dependence upon subject heterogeneity for existence of meaningful relationships.

Although sprinting and the HT exercise share some common biomechanical features pertaining to sagittally oriented force originating predominantly from the hip-extensor musculature, which could explain previous reports of a robust relationships between these variables (Loturco et al., 2018; Williams et al., 2021), they are both associated with unique kinematical characteristics potentially underlying their apparent distinctiveness as presently observed. During the push-off phase of sprinting when horizontal ground impulse is transmitted onto the supporting ground, the knee-joint operates in relative proximity of terminal extension(Nagahara et al., 2014), whereas the operating point of the knee in the HT is in a substantially more flexed position(Contreras et al., 2011). The relative importance of the various hip-extensor muscles to hip-extension torque production is known to vary as a function of knee-joint angle, with the bi-articular hamstrings being of stronger importance when the knee is closer to full extension and with the uni-articular gluteals serving as the prime hip-extensor in a knee-flexed position (Kim & Park, 2016; Kwon & Lee, 2013).

Limitations

A number of potential limitations may be noted with the present study. Although the F-v relationship of the HT was relatively well-fitted by linear regression, knowledge regarding the validity of the derived neuromuscular capabilities is scarce. Whereas the established FVP profiling modalities of sprinting and jumping are associated with maximal effort body mass accelerations, a ballistic propulsion of the barbell is a necessity in the HT to avoid an unwanted voluntary deceleration towards the terminal range of motion. Although speculative, it is possible that this style of execution, without spotters to subsequently catch the barbell, provided a psychological impedance hindering provision of a true maximal effort, thus potentially compromising construct validity. Additionally, it should be recognized that although the high-frequency video analysis utilized for the HT is a validated methodology for the quantification of mean concentric velocity(Balsalobre-Fernandez et al., 2017, 2018), the associated mathematical integration does not account for the brief isometric contraction preceding the concentric phase, thus potentially confounding the results. Regarding the sprint-specific neuromuscular capabilities, these were computed via a time-position function obtained through split times, a methodology known to be associated with a systematic bias in comparison to methodologies utilizing direct quantification of the time-velocity curve (Samozino et al., 2016), e.g., radar/laser technologies. However, to avoid/minimize this bias, the recently validated time-shift methodology (Morin et al., 2019; Stenroth et al., 2020) was applied, and though information regarding the external validity of this methodology is scarce, the remarkably similar neuromuscular capabilities compared to previous observations in elite male footballers(Jimenez-Reyes et al., 2019; 2018), could be indicative of successful circumvention of this known bias. Finally, the participant population encompassed a homogenous group of elite football players and the present results may thus not necessarily be extrapolatable to other athletic populations or untrained individuals.

CONCLUSIONS

In conclusion, although Pmax may be considered a somewhat universal lower-extremity neuromuscular capability, the main inference of the present study is the marked task-specificity of its underlying mechanical determinants, i.e. F0 and V0. Furthermore, the present data suggest task-specificity to be a complex multifactorial phenomenon not exclusively governed by the overall orientation of force application, likely encompassing both force orientation specificity and dynamic correspondence. The present findings are not necessarily in conflict with the theory of force orientation specificity for training purposes or disprove the use of the HT exercise for improving sprint acceleration capacity, however, the distinctiveness of their respective neuromuscular capabilities may suggest a limited transference from the HT exercise into sprint-specific improvements in neuromuscular function for

high-level athletes. Although isolated hip-extensor neuromuscular function appears distinct from that of sprint running, implementation of HT FVP profiling may still provide additional comprehensiveness regarding individual constraints to sprint capacity to be targeted through individualized training.

Acknowledgments

Nothing to declare.

Disclosure statement

Bret Contreras is the inventor of the hip thrust exercise.

ORCID

Nicklas Junge http://orcid.org/0000-0002-3874-1870
Pierre Samozino http://orcid.org/0000-0002-1665-870X
Jean-Benoit Morin http://orcid.org/0000-0003-3808-6762
Lars Nybo http://orcid.org/0000-0002-9090-1958

References

Abade, E., Silva, N., Ferreira, R., et al. (2019). Effects of adding vertical or horizontal force-vector exercises to in-season general strength training on jumping and sprinting performance of youth football players. *Journal of Strength and Conditioning Research*, *Publish Ahead of Print*,https://doi.org/10.1519/JSC.0000000000003221

Alonso-Aubin, D. A., Chulvi-Medrano, I., Cortell-Tormo, J. M., Picon-Martinez, M., Rial Rebullido, T., & Faigenbaum, A. D. (2019). Squat and bench press force-velocity profiling in male and female adolescent rugby players. *Journal Of Strength And Conditioning Research / National Strength & Conditioning Association*.

Balsalobre-Fernandez, C., Marchante, D., Baz-Valle, E., Alonso-Molero, I., Jimenez, S. L., & Munoz-Lopez, M. (2017). Analysis of wearable and smartphone-based technologies for the measurement of barbell velocity in different resistance training exercises. *Frontiers in Physiology*, *8*, 649. https://doi.org/10.3389/fphys.2017.00649

Balsalobre-Fernandez, C., Marchante, D., Munoz-Lopez, M., & Jimenez, S. L. (2018). Validity and reliability of a novel iPhone app for the measurement of barbell velocity and 1RM on the bench-press exercise. *Journal of Sports Sciences*, *36*(1), 64–70. https://doi.org/10.1080/02640414.2017.1280610

Cometti, G., Maffiuletti, N. A., Pousson, M., Chatard, J. C., & Maffulli, N. (2001). Isokinetic strength and anaerobic power of elite, subelite and amateur French soccer players. *International Journal of Sports Medicine*, *22*(1), 45–51. https://doi.org/10.1055/s-2001-11331

Contreras, B., Cronin, J., & Schoenfeld, B. (2011). Barbell Hip Thrust. *33*, 58–61.

Contreras, B., Vigotsky, A. D., Schoenfeld, B. J., et al. (2017). Effects of a six-week hip thrust vs. front squat resistance training program on performance in adolescent males: a randomized controlled trial. *Journal of Strength and Conditioning Research*, *31*(4), 999–1008. https://doi.org/10.1519/JSC.0000000000001510

Contreras, B., Vigotsky, A. D., Schoenfeld, B. J., Beardsley, C., & Cronin, J. (2015). A comparison of gluteus maximus, biceps femoris, and vastus lateralis electromyographic activity in the back squat and barbell hip thrust exercises. *Journal of Applied Biomechanics*, *31*(6), 452–458. https://doi.org/10.1123/jab.2014-0301

Escobar-Alvarez, J. A., Fuentes-García, J., Viana-da-conceicao, F. A., & Jiménez-Reyes, P. (2020). Association between vertical and horizontal force-velocity-power profiles in netball players. Journal of Human Sport and Exercise, 17(1). https://doi.org/10.14198/jhse.2022.171.09

Faude, O., Koch, T., & Meyer, T. (2012). Straight sprinting is the most frequent action in goal situations in professional football. *Journal of Sports Sciences*, *30*(7), 625–631. https://doi.org/10.1080/02640414.2012.665940

Garcia-Ramos, A., Jaric, S., Padial, P., & Force-Velocity, F. B. (2016). Relationship of upper body muscles: traditional versus ballistic bench press. *Journal of Applied Biomechanics*, *32*(2), 178–185. https://doi.org/10.1123/jab.2015-0162

Girard, O., Brocherie, F., Morin, J. B., Millet, G. P., & Hansen, C. (2020). Running mechanics and leg muscle activity patterns during early and late acceleration phases of repeated treadmill sprints in male recreational athletes. *European Journal of Applied Physiology*, *120*(12), 2785–2796. https://doi.org/10.1007/s00421-020-04500-8

Glatthorn, J. F., Gouge, S., Nussbaumer, S., Stauffacher, S., Impellizzeri, F. M., & Maffiuletti, N. A. (2011). Validity and reliability of Optojump photoelectric cells for estimating vertical jump height. *Journal of Strength and Conditioning Research*, *25*(2), 556–560. https://doi.org/10.1519/JSC.0b013e3181ccb18d

Gonzalez-Garcia, J., Morencos, E., Balsalobre-Fernandez, C., Cuellar-Rayo, A., & Romero-Moraleda, B. (2019). Effects of 7-week hip thrust versus back squat resistance training on performance in adolescent female soccer players. *Sports (Basel, Switzerland)*, *7*(4), 7. https://doi.org/10.3390/sports7040080

Haugen, T. A., Breitschadel, F., & Samozino, P. (2020). Power-force-velocity profiling of sprinting athletes: methodological and practical considerations when using timing gates. *Journal of Strength and Conditioning Research*, *34*(6), 1769–1773. https://doi.org/10.1519/JSC.0000000000002890

Jaric, S. (2015). Force-velocity relationship of muscles performing multi-joint maximum performance tasks. *International Journal of Sports Medicine*, *36*(9), 699–704. https://doi.org/10.1055/s-0035-1547283

Jarvis, P., Cassone, N., Turner, A., Chavda, S., Edwards, M., & Bishop, C. (2019). Heavy barbell hip thrusts do not effect sprint performance: an 8-week randomized controlled study. *Journal of Strength and Conditioning Research*, *33*(Suppl 1), S78–S84. https://doi.org/10.1519/JSC.0000000000002146

Jimenez-Reyes, P., Garcia-Ramos, A., Cuadrado-Penafiel, V., et al. (2019). Differences in sprint mechanical force-velocity profile between trained soccer and futsal players. *International Journal of Sports Physiology and Performance*, *14*(4), 478–485. https://doi.org/10.1123/ijspp.2018-0402

Jimenez-Reyes, P., Samozino, P., Garcia-Ramos, A., Cuadrado-Penafiel, V., Brughelli, M., & Morin, J. B. (2018). Relationship between vertical and horizontal force-velocity-power profiles in various sports and levels of practice. *PeerJ*, *6*, e5937. https://doi.org/10.7717/peerj.5937

Kim, J., & Park, M. (2016). Changes in the activity of trunk and hip extensor muscles during bridge exercises with variations in unilateral knee joint angle. *Journal of Physical Therapy Science*, *28*(9), 2537–2540. https://doi.org/10.1589/jpts.28.2537

Kwon, Y.-J., & Lee, H.-O. (2013). How different knee flexion angles influence the hip extensor in the prone position. *Journal of Physical Therapy Science*, *25*(10), 1295–1297. https://doi.org/10.1589/jpts.25.1295

Loturco, I., Contreras, B., Kobal, R., et al. (2018). Vertically and horizontally directed muscle power exercises: Relationships with top-level sprint performance. *PLoS One*, *13*(7), e0201475. https://doi.org/10.1371/journal.pone.0201475

Marcote-Pequeno, R., Garcia-Ramos, A., Cuadrado-Penafiel, V., Gonzalez-Hernandez, J. M., Gomez, M. A., & Association, J.-R. P. (2019). Between the force-velocity profile and performance variables obtained in jumping and sprinting in elite female soccer players. *International Journal of Sports Physiology and Performance*, *14*(2), 209–215. https://doi.org/10.1123/ijspp.2018-0233

Morin, J. B., Edouard, P., & Samozino, P. (2011). Technical ability of force application as a determinant factor of sprint performance. *Medicine & Science in Sports & Exercise*, *43*(9), 1680–1688. https://doi.org/10.1249/MSS.0b013e318216ea37

Morin, J. B., Gimenez, P., Edouard, P., et al. (2015). Sprint Acceleration Mechanics: The Major Role of Hamstrings in Horizontal Force Production. *Frontiers in Physiology*, *6*, 404. https://doi.org/10.3389/fphys.2015.00404

Morin, J. B., Jimenez-Reyes, P., Brughelli, M., & Samozino, P. (2019). When jump height is not a good indicator of lower limb maximal power output: theoretical demonstration, experimental evidence and practical solutions. *Sports Medicine*, *49*(7), 999–1006. https://doi.org/10.1007/s40279-019-01073-1

Morin, J.-B., Samozino, P., Murata, M., Cross, M. R., & Nagahara, R. (2019). A simple method for computing sprint acceleration kinetics from running velocity data: Replication study with improved design. *Journal of Biomechanics*, *94*, 82–87. https://doi.org/10.1016/j.jbiomech.2019.07.020

Nagahara, R., Matsubayashi, T., Matsuo, A., & Zushi, K. (2014). Kinematics of transition during human accelerated sprinting. *Biology Open*, *3*(8), 689–699. https://doi.org/10.1242/bio.20148284

Padulo, J., Migliaccio, G. M., Ardigo, L. P., Leban, B., Cosso, M., & Samozino, P. (2017). Lower limb force, velocity, power capabilities during leg press and squat movements. *International Journal of Sports Medicine*, *38*(14), 1083–1089. https://doi.org/10.1055/s-0043-118341

Rabita, G., Dorel, S., Slawinski, J., et al. (2015). Sprint mechanics in world-class athletes: A new insight into the limits of human locomotion. *Scandinavian Journal of Medicine & Science in Sports*, *25*(5), 583–594. https://doi.org/10.1111/sms.12389

Rahmani, A., Samozino, P., Morin, J. B., & Morel, B. (2018). A simple method for assessing upper-limb force-velocity profile in bench press. *International Journal of Sports Physiology and Performance*, *13*(2), 200–207. https://doi.org/10.1123/ijspp.2016-0814

Rambaud, O., Rahmani, A., Moyen, B., & Bourdin, M. (2008). Importance of upper-limb inertia in calculating concentric bench press force. *Journal of Strength and Conditioning Research*, *22*(2), 383–389. https://doi.org/10.1519/JSC.0b013e31816193e7

Samozino, P., Edouard, P., Sangnier, S., Brughelli, M., Gimenez, P., & Morin, J. B. (2014). Force-velocity profile: Imbalance determination and effect on lower limb ballistic performance. *International Journal of Sports medicine*, *35*(6), 505–510. https://doi.org/10.1055/s-0033-1354382

Samozino, P., Morin, J. B., Hintzy, F., & Belli, A. (2008). A simple method for measuring force, velocity and power output during squat jump. *Journal of Biomechanics*, *41*(14), 2940–2945. https://doi.org/10.1016/j.jbiomech.2008.07.028

Samozino, P., Rabita, G., Dorel, S., et al. (2016). A simple method for measuring power, force, velocity properties, and mechanical effectiveness in sprint running. *Scandinavian Journal of Medicine & Science in Sports*, *26*, 648–658. https://doi.org/10.1111/sms.12490

Sreckovic, S., Cuk, I., Djuric, S., Nedeljkovic, A., Mirkov, D., & Jaric, S. (2015). Evaluation of force-velocity and power-velocity relationship of arm muscles. *European Journal of Applied Physiology*, *115*(8), 1779–1787. https://doi.org/10.1007/s00421-015-3165-1

Stenroth, L., Vartiainen, P., & Karjalainen, P. A. (2020). Force-velocity profiling in ice hockey skating: Reliability and validity of a simple, low-cost field method. *Sports Biomechanics*, 1–16. https://doi.org/10.1080/14763141.2020.1770321

Wiemann, K., & Tidow, G. (1995). Relative activity of hip and knee extensors in sprinting - implications for training. In *New studies in athletics* (pp. 29–49).

Williams, M. J., Gibson, N. V., Sorbie, G. G., Ugbolue, U. C., Brouner, J., & Easton, C. (2021). Activation of the Gluteus Maximus During Performance of the Back Squat, Split Squat, and Barbell Hip Thrust and the Relationship With Maximal Sprinting. *Journal of Strength and Conditioning Research*, *35*(1), 16–24. https://doi.org/10.1519/JSC.0000000000002651